

TEN Controller v1.0

Technical Documentation

4-channel independent heater control system
ESP-32 DevKit1 | K-thermocouples | Current sensing

Parameter	Value
MCU	ESP-32 DevKit1
Heaters	4 x 1 kW, 220V AC
Temperature sensor	K-type thermocouple + MAX6675
Current sensing	Split-core CT + conditioning circuit
Control range	435-450 C (hysteresis)
Indication	12 LEDs (Yellow/Green/Red) via 74HC595
Serial interface	115200 baud (STATUS, RESET, STOP, CAL)
GPIO used	18 pins total

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1. System Overview

The system controls four independent 1 kW heaters powered by 220V AC mains. Each channel has its own K-type thermocouple for temperature measurement via MAX6675 module, a split-core current transformer (CT) for verifying heater current flow, and a power relay for switching. All channels are managed by a single ESP-32 DevKit1 microcontroller. Temperature regulation uses hysteresis control with turn-off threshold at 450 degrees Celsius and turn-on threshold at 435 degrees Celsius, providing a 15-degree deadband to prevent relay chatter and ensure stable operation.

Visual indication is provided by 12 LEDs arranged in groups of three per channel: yellow for active heating, green for temperature within the regulation range, and red for fault detection (heater burned out or open circuit). Two 74HC595 shift registers cascaded together drive all LEDs using only three ESP-32 GPIO pins, which is critical given the limited pin budget of the controller.

Системная блок-схема ТЭН Controller v1.0

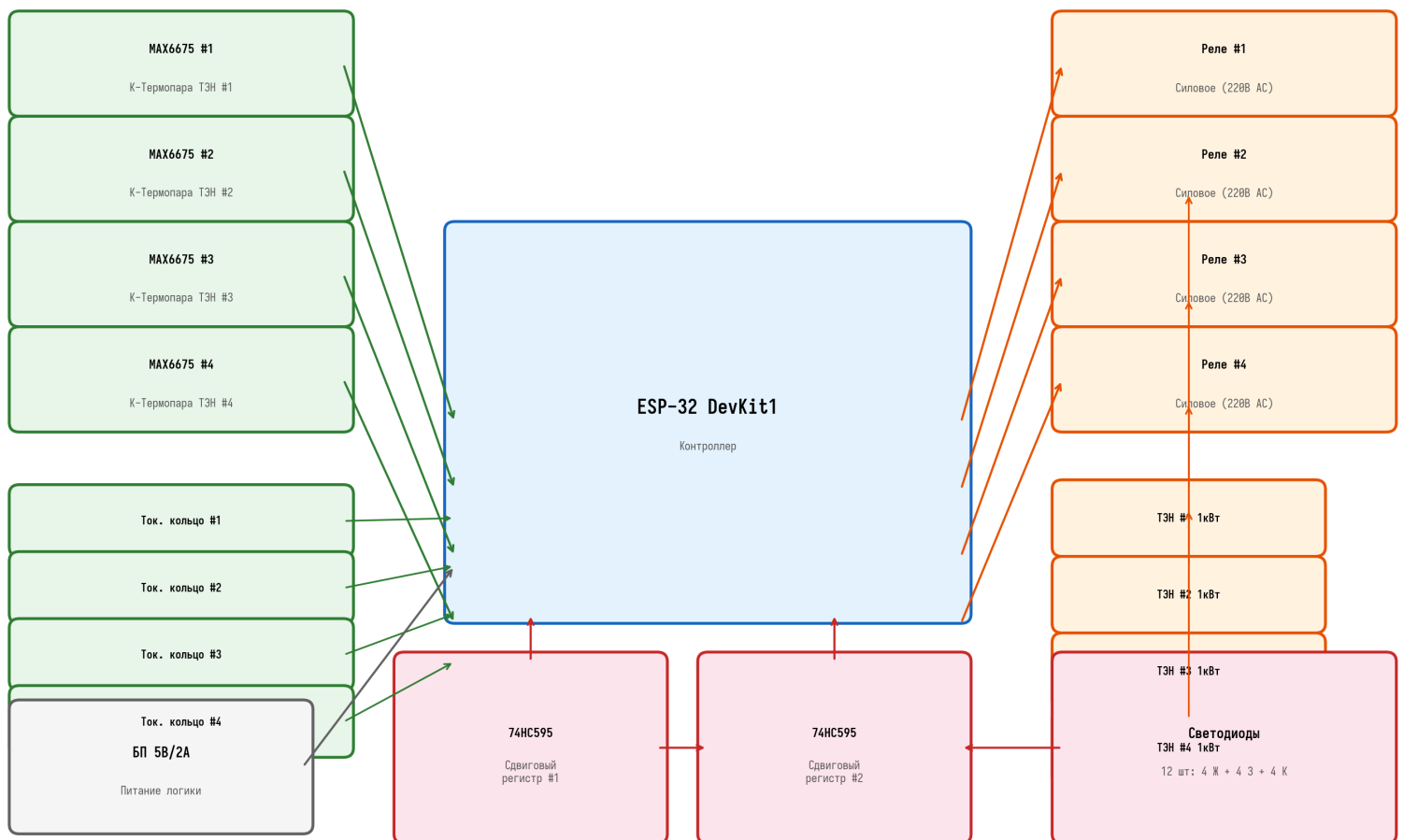


Fig. 1. System block diagram of TЭН Controller v1.0

2. Power Circuit (220V AC)

Each heater channel has an identical power circuit. The 220V AC phase wire passes through a split-core current transformer before reaching the power relay contacts. The relay then connects the phase wire to one terminal of the heating element. The neutral wire connects

directly to the other terminal of the heating element. The current transformer secondary winding produces a small AC current proportional to the load current (typically 1mA per ampere of primary current for SCT-013-000), which is then conditioned and fed to the ESP-32 ADC for monitoring.

The power relay is controlled through a low-voltage relay module with optocoupler isolation. This is critically important for safety: the optocoupler provides galvanic isolation between the low-voltage ESP-32 circuitry and the 220V AC mains, protecting the microcontroller and the user from high voltage in case of a relay failure. The relay module is active-LOW, meaning that setting the GPIO pin to LOW closes the relay contacts and applies power to the heater.

WARNING: 220V AC is lethal! All power connections must be enclosed in an insulated enclosure. Use proper wire gauges (minimum 1.5 mm squared for 4.5A per channel). Always disconnect power before making any modifications. An external 16A circuit breaker is mandatory for the entire system.

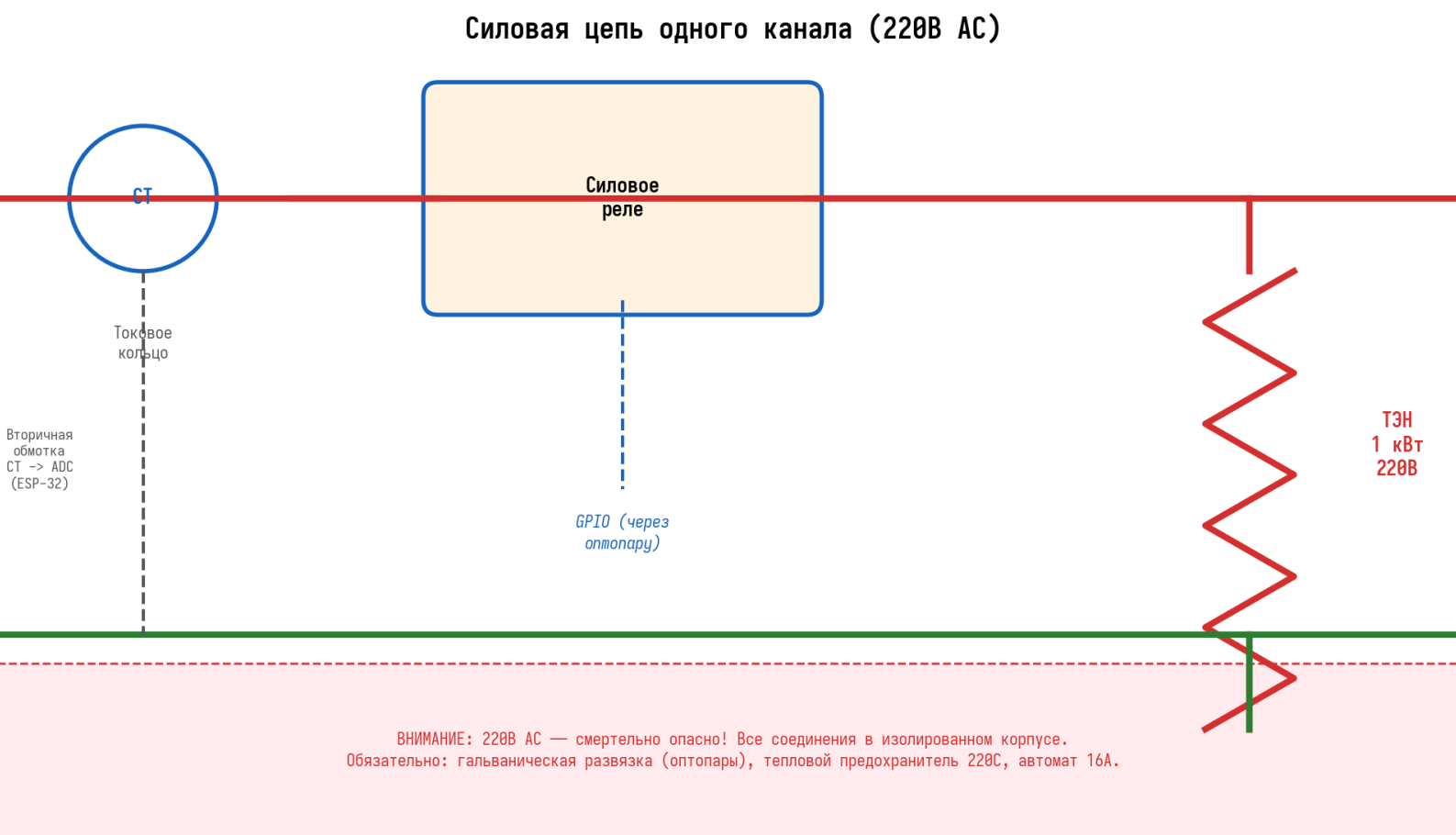


Fig. 2. Single channel power circuit diagram (220V AC)

3. MAX6675 Thermocouple Wiring

Temperature measurement uses MAX6675 modules, each of which includes a cold-junction compensator and a 12-bit ADC specifically designed for K-type thermocouples. The measurement range extends from 0 to 1024 degrees Celsius with a resolution of 0.25 degrees per LSB. The

module communicates via a simple SPI-like protocol using three wires: SCK (clock), SO (data output), and CS (chip select).

All four MAX6675 modules share the same SCK and SO lines, forming a single bus. Each module has its own dedicated CS pin on the ESP-32, allowing individual selection during the polling sequence. The bit-bang SPI implementation in the firmware manually toggles the clock and reads data bits, which avoids conflicts with the ESP-32 hardware SPI peripheral that may be needed for other purposes. A minimum delay of 250 ms between consecutive readings of the same module is required for the ADC conversion to complete.

Note: MAX6675 modules are typically rated for 3.3V to 5V supply. Check your specific module datasheet before connecting. The K-thermocouple wires must be connected with correct polarity: yellow wire to T+ (chromel) and red wire to T- (alumel). Reversed polarity will result in negative temperature readings.

Fig. 3. MAX6675 module wiring diagram (4 modules on shared SPI bus)

4. Current Sensor Conditioning Circuit

The split-core current transformer (CT) produces an AC current on its secondary winding proportional to the current flowing through the primary conductor (the heater power wire). For the SCT-013-000 model with a 2000:1 turns ratio, a 4.5A heater current produces approximately 2.25 mA secondary current. This AC signal cannot be read directly by the ESP-32 ADC and must be conditioned through the following stages.

First, a ballast resistor (R_b , 100 Ω) converts the secondary current to a small AC voltage (approximately 225 mV peak for nominal heater current). Second, a diode bridge rectifies this AC voltage to DC. Third, a filter capacitor (C_f , 100 μ F) smooths the rectified signal to reduce ripple from the 50 Hz mains frequency. Fourth, a potentiometer (R_{adj} , 10 k Ω) allows fine-tuning of the signal amplitude to match the ADC input range. Finally, a pull-down resistor (R_{gnd} , 10 k Ω) provides a defined output when the CT secondary is open-circuit, preventing floating ADC readings.

After conditioning, the expected ADC reading for a healthy 1 kW heater at 220V is approximately 248 units (on a 12-bit 0-4095 scale with 3.3V reference). The default CURRENT_THRESH of 100 units provides adequate margin below this value while still being significantly above the noise floor. During calibration, the actual ADC value should be measured and the threshold adjusted to 50-70% of the observed working value for reliable fault detection.

Цепь токового кольца (Conditioning Circuit)

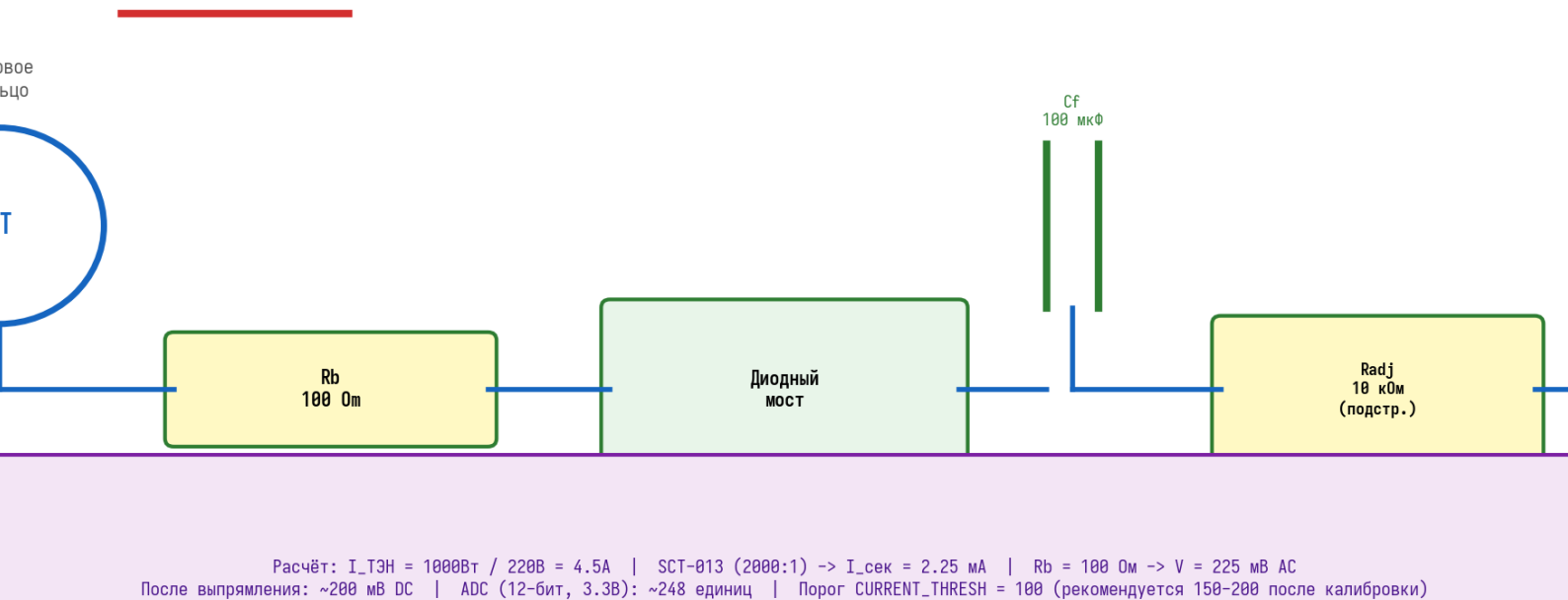


Fig. 4. Current sensor conditioning circuit with calculations

5. 74HC595 Shift Register Cascade

The LED indication system uses 12 LEDs (3 per channel) but the ESP-32 has limited GPIO pins available after assigning pins for thermocouples, relays, and current sensors. The 74HC595 shift register solves this problem by controlling up to 8 outputs using only 3 GPIO pins. Two registers are cascaded to provide 16 output channels (of which 12 are used for the LEDs).

The cascade connection works by routing the Q7 prime (serial output) of the first register to the DS (serial input) of the second register. Both chips share the same SHCP (shift clock) and STCP (storage clock) signals. During operation, the firmware shifts out a 16-bit pattern where the lower 12 bits correspond to individual LEDs and the upper 4 bits are unused. The bit assignment follows a repeating pattern of Yellow, Green, Red for each channel: bits 0-2 for heater 1, bits 3-5 for heater 2, bits 6-8 for heater 3, and bits 9-11 for heater 4.

The OE (output enable) pin of both registers should be connected to ground for continuous output, and the SRCLR (master reset) pin should be tied to 5V to prevent accidental clearing. Each LED requires a series current-limiting resistor (330 Ohm is standard for 5V supply) to prevent excessive current through the shift register outputs.

Каскад 74HC595: управление 12 светодиодами

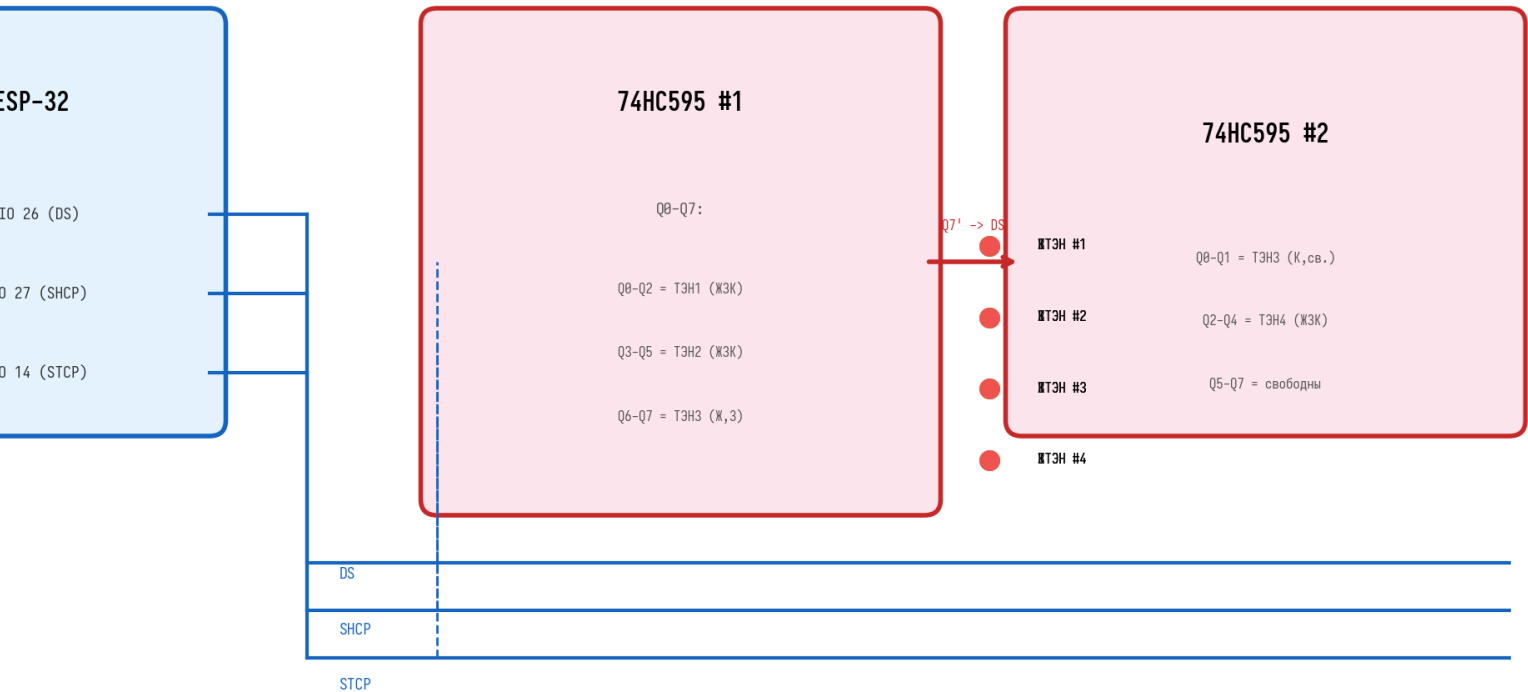


Fig. 5. 74HC595 cascade wiring for 12 LED indicators

6. ESP-32 Pin Assignment

The ESP-32 DevKit1 provides 34 GPIO pins, of which 18 are used in this design. The pin selection follows specific constraints: thermocouple CS pins and relay control pins must be standard bidirectional GPIOs; current sensor inputs must be ADC1 channels (GPIOs 34-39 are input-only and support ADC1, while ADC2 cannot be used when WiFi is active); and shift register pins must be standard outputs. The following table shows the complete pin assignment.

Полная распиновка ESP-32 DevKit1

Пин ESP-32	Тип	Назначение	Примечание
GPIO 18	OUT	MAX6675 SCK	Общий тактовый сигнал SPI
GPIO 19	IN	MAX6675 S0 (MISO)	Общие данные SPI
GPIO 5	OUT	MAX6675 CS #1	Термопара ТЭН #1
GPIO 4	OUT	MAX6675 CS #2	Термопара ТЭН #2
GPIO 16	OUT	MAX6675 CS #3	Термопара ТЭН #3
GPIO 17	OUT	MAX6675 CS #4	Термопара ТЭН #4
GPIO 23	OUT	Реле ТЭН #1	Active-LOW (LOW = ВКЛ)
GPIO 22	OUT	Реле ТЭН #2	Active-LOW (LOW = ВКЛ)
GPIO 21	OUT	Реле ТЭН #3	Active-LOW (LOW = ВКЛ)
GPIO 13	OUT	Реле ТЭН #4	Active-LOW (LOW = ВКЛ)
GPIO 36	IN	ADC Ток. кольцо #1	ADC1_CH0 (input-only)
GPIO 39	IN	ADC Ток. кольцо #2	ADC1_CH3 (input-only)
GPIO 34	IN	ADC Ток. кольцо #3	ADC1_CH6 (input-only)
GPIO 35	IN	ADC Ток. кольцо #4	ADC1_CH7 (input-only)
GPIO 26	OUT	74HC595 DS	Данные сдвигового регистра
GPIO 27	OUT	74HC595 SHCP	Такты сдвигового регистра
GPIO 14	OUT	74HC595 STCP	Защёлка сдвигового регистра
3.3V	PWR	MAX6675 VCC	Питание модулей термопар
5V	PWR	74HC595 VCC + Реле	Через внешний БП 5В/2А
GND	GND	Обий земля	Все компоненты

Fig. 6. Complete ESP-32 DevKit1 pin assignment table

GPIO 18	OUT	MAX6675 SCK	Shared SPI clock for all thermocouples
GPIO 19	IN	MAX6675 S0	Shared SPI data for all thermocouples
GPIO 5	OUT	MAX6675 CS #1	Chip select - Heater 1 thermocouple
GPIO 4	OUT	MAX6675 CS #2	Chip select - Heater 2 thermocouple
GPIO 16	OUT	MAX6675 CS #3	Chip select - Heater 3 thermocouple
GPIO 17	OUT	MAX6675 CS #4	Chip select - Heater 4 thermocouple
GPIO 23	OUT	Relay #1	Active-LOW: LOW = heater ON
GPIO 22	OUT	Relay #2	Active-LOW: LOW = heater ON
GPIO 21	OUT	Relay #3	Active-LOW: LOW = heater ON
GPIO 13	OUT	Relay #4	Active-LOW: LOW = heater ON
GPIO 36	IN	Current ADC #1	ADC1_CH0, input-only pin
GPIO 39	IN	Current ADC #2	ADC1_CH3, input-only pin
GPIO 34	IN	Current ADC #3	ADC1_CH6, input-only pin
GPIO 35	IN	Current ADC #4	ADC1_CH7, input-only pin
GPIO 26	OUT	74HC595 DS	Serial data to shift registers

GPIO 27	OUT	74HC595 SHCP	Shift register clock
GPIO 14	OUT	74HC595 STCP	Storage register latch

7. State Machine

Each heater channel is controlled by an independent finite state machine (FSM) with six states. The FSM handles temperature regulation, fault detection, and error recovery. The transitions between states are determined by the thermocouple reading and the current sensor value, with specific timing constraints to prevent false fault detection.

State Descriptions

IDLE (Waiting) - Initial state after power-on or reset. The relay is OFF and the heater is de-energized. The system waits for the temperature to drop below 435 degrees before enabling heating. If the temperature is already below this threshold, the FSM transitions directly to HEATING.

HEATING (Active) - The relay is ON and the heater is energized. The yellow LED is illuminated. The current sensor is monitored after a 3-second delay (to allow relay contact settling). If the current drops below the threshold, a FAULT condition is triggered immediately. Heating continues until the temperature reaches 450 degrees.

IN_RANGE (Regulation) - Temperature is between 435 and 450 degrees with the relay OFF. The green LED indicates that the system is actively regulating. The heater waits for the temperature to drop below 435 before the next heating cycle begins.

COOLING (Above Threshold) - Temperature is above 450 degrees. The relay has been turned off. The system waits for natural cooling or heat dissipation until the temperature drops below 450, at which point it transitions to IN_RANGE.

FAULT (Burned Out) - Critical error detected: the relay is ON but no current is flowing through the heater. This indicates a burned-out heating element, an open circuit, or a relay contact failure. The red LED is illuminated and the fault is latched (persistent) until manually cleared via the RESET serial command. The relay is immediately de-energized for safety.

THERMO_ERR (Sensor Error) - The MAX6675 has reported five consecutive invalid readings (thermocouple open or disconnected). The red LED blinks at approximately 1.7 Hz. The relay is turned off. This state auto-clears when valid readings resume.

Конечный автомат управления каналом ТЭН

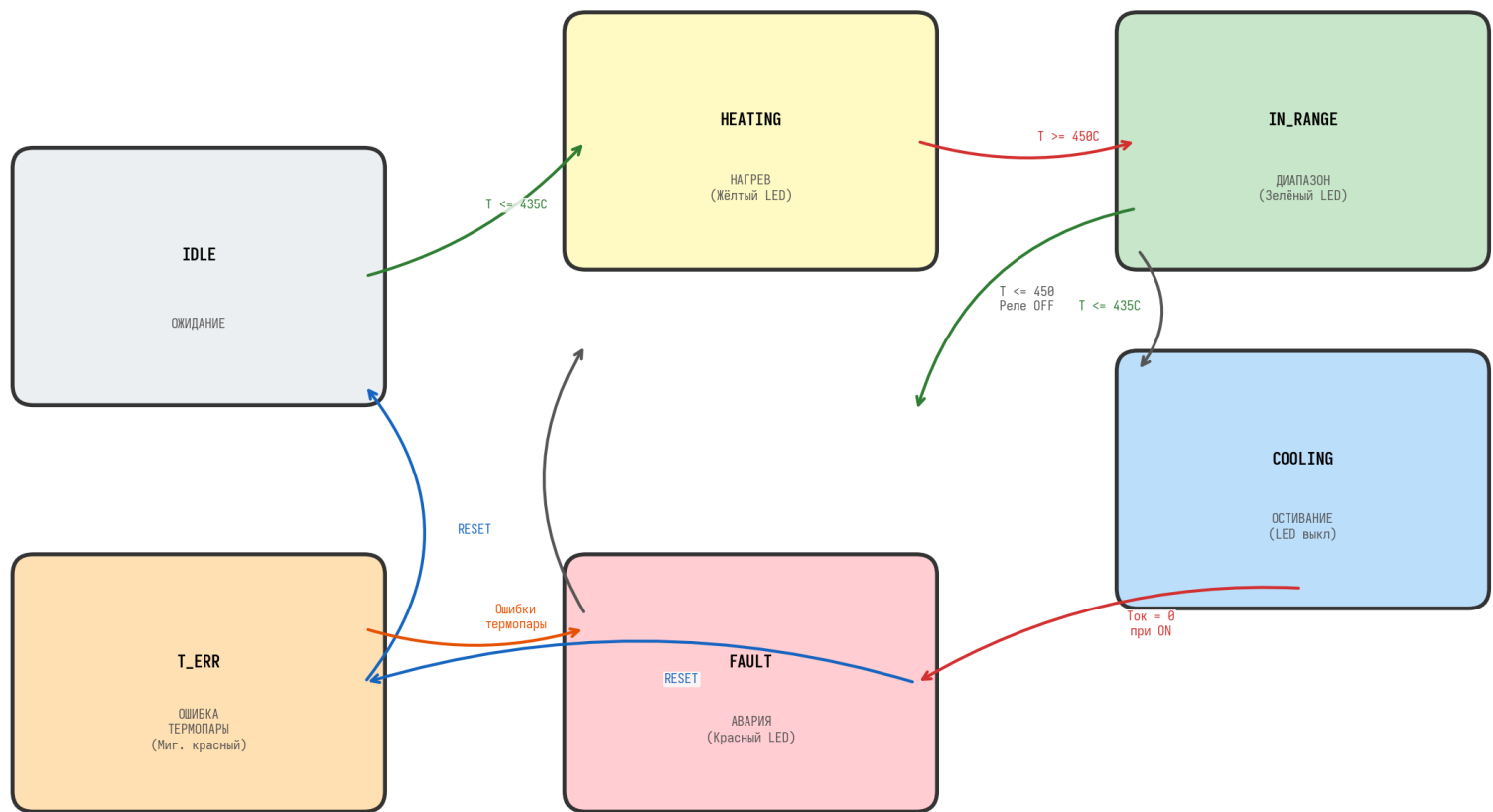


Fig. 7. Finite state machine diagram for single heater channel

8. Calibration Procedure

Current sensor calibration is essential for reliable fault detection. The `CURRENT_THRESH` parameter in the firmware must be set to a value that reliably distinguishes between a healthy heater (drawing approximately 4.5A) and a burned-out heater (drawing 0A). The default value of 100 ADC units may need adjustment depending on the specific current transformer model, burden resistor value, and potentiometer setting.

Step-by-Step Calibration

Step 1: Upload the firmware to ESP-32 and connect to Serial Monitor at 115200 baud rate. Verify that all thermocouples are reading correctly (no ERR status in the periodic status output).

Step 2: Send the command "CAL 1" to start calibration for heater channel 1. The system will energize the relay and display 10 consecutive ADC readings at 500ms intervals. Observe the readings and note the stable average value.

Step 3: Calculate the threshold as 50-70% of the observed ADC value. For example, if the readings stabilize around 350, set `CURRENT_THRESH` to 175-245. A value of 200 provides a good safety margin above noise (typically 5-20 ADC units) while still detecting open-circuit conditions.

Step 4: Edit the CURRENT_THRESH constant in the firmware source code and re-upload. Repeat the calibration procedure for all four channels, as minor variations in CT characteristics and wiring resistance may cause different ADC readings per channel.

Процедура калибровки порога тока



Fig. 8. Calibration procedure flowchart

9. Component List (Bill of Materials)

The following table lists all components required to build one complete controller unit with four independent heater channels. All components are commonly available from electronics suppliers. Prices are approximate and may vary by region and supplier.

Qty	Component	Specification	Purpose
1	ESP-32 DevKit1	38-pin, WiFi+BT	Main controller MCU
4	Heating element	1 kW, 220V AC	Load to be controlled
4	MAX6675 module	K-thermocouple amplifier	Temperature measurement
4	K-type thermocouple	Probe, up to 500C	Temperature sensor
4	Split-core CT	SCT-013-000, 100A/50mA	Current monitoring
4	Power relay	10A+, 250VAC coil	Heater power switching
4	Relay module	5V, optocoupler, active-LOW	Logic-level relay driver
4	Diode bridge	W04M or equivalent	CT signal rectification

4	Resistor 100 Ohm	0.5W, 1%	CT burden resistor (Rb)
4	Capacitor 100 uF	16V electrolytic	CT signal filtering (Cf)
4	Trimmer 10 kOhm	Through-hole	CT signal calibration (Radj)
4	Resistor 10 kOhm	0.25W, 1%	ADC pull-down (Rgnd)
2	74HC595	DIP-16	LED shift register
4	LED Yellow	5mm, diffused	Heating indicator
4	LED Green	5mm, diffused	In-range indicator
4	LED Red	5mm, diffused	Fault indicator
12	Resistor 330 Ohm	0.25W	LED current limiting
1	Power supply	5V, 2A, USB or barrel	Logic power
1	Thermal fuse	220C, 10A	Safety cutoff (per heater)
1	Enclosure	IP54 rated	Safety and protection

10. Serial Commands Reference

The firmware provides a simple command-line interface via the USB serial connection at 115200 baud. Commands are used for diagnostics, fault recovery, emergency shutdown, and calibration. Commands are not case-sensitive and are terminated by a newline character.

Command	Description	Example
STATUS	Print status of all 4 channels	STATUS
RESET n	Clear fault on channel n (1-4)	RESET 2
STOP n	Emergency disable channel n	STOP 1
STOP ALL	Emergency disable all channels	STOP ALL
CAL n	Start current calibration on ch n	CAL 3
THRESH v	Show threshold info (edit in code)	THRESH 200

Status Output Format

```
===== STATYC =====
ТЭН #1 | 437.2C | Ток: 347 | Реле: ВКЛ | НАГРЕВ | [ЖЁЛТ]
ТЭН #2 | 442.1C | Ток: 0 | Реле: ВЫКЛ | ДИАПАЗОН | [ЗЕЛЁН]
ТЭН #3 | 451.0C | Ток: 0 | Реле: ВЫКЛ | ОСТИВАНИЕ | [---]
ТЭН #4 | ERR | Ток: 0 | Реле: ВЫКЛ | АВАРИЯ !!! | [КРАСН]
=====
```

The status output is printed automatically every 2 seconds and shows for each channel: the current thermocouple temperature (or ERR if reading failed), the raw ADC value from the current sensor, the relay state, the FSM state name, and the active LED color. This provides comprehensive real-time visibility into the system operation for debugging and monitoring purposes.

11. Safety Notes

ELECTRICAL SAFETY: This project involves 220V AC mains voltage which is potentially lethal. Only qualified personnel should assemble, wire, and test this system. Always disconnect power before making any changes to the wiring. Use insulated tools and wear appropriate PPE.

Galvanic Isolation: The relay modules with optocouplers provide essential isolation between the low-voltage ESP-32 circuit (3.3V-5V) and the 220V AC mains. Never bypass the optocouplers or connect mains voltage directly to ESP-32 pins. The current transformer also provides inherent isolation between the power circuit and the measurement circuit.

Thermal Protection: Each heater channel should have a thermal fuse rated at 220 degrees Celsius installed in series with the heating element as a last-resort safety device. This mechanical fuse will permanently open the circuit if the electronic control fails and the temperature exceeds the safe operating range.

Overcurrent Protection: A dedicated 16A circuit breaker should protect the entire system. With four heaters at 1 kW each (approximately 4.5A each), the total system current is approximately 18A at full load. Individual fusing per channel is recommended but the shared circuit breaker provides basic protection against short circuits.

Safe Startup: The firmware initializes all relay control pins to HIGH (relay OFF) at the very beginning of the setup() function, before any other initialization. This ensures that heaters remain de-energized even if the ESP-32 resets unexpectedly. The heaters will only activate after the thermocouples have been verified and the temperature is confirmed to be below the 435 degree threshold.

Enclosure: All high-voltage connections must be enclosed in a properly rated electrical enclosure (minimum IP54 rating). The enclosure should prevent accidental contact with live parts and protect against dust and moisture ingress. Ventilation may be required depending on the operating environment and the heat dissipation characteristics of the enclosure.